



DATA SCIENCE MINI PROJECT

Study Habits and Academic Performance of Engineering Students During Holidays

An exploratory survey-based analysis of how engineering students utilise their holiday time, and how habits such as study routines, attendance, and digital-tool usage relate to academic performance.

Survey Sample: 20 Engineering Students (Google Forms Survey)

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Current CGPA: 9.5

Introduction & Objectives

Holidays form a significant portion of an academic year, yet how students choose to spend this time varies widely — from intensive self-study to complete disengagement. This project uses survey data collected from 20 engineering students to explore patterns in holiday study behaviour, attendance attitudes, AI-tool adoption, and the challenges students face, in order to draw data-driven insights about what supports (or hinders) academic performance.

Survey Method: An anonymous Google Forms questionnaire covering branch, study hours, CGPA, attendance attitudes, holiday activities, AI-tool usage, and study challenges was circulated among engineering students, receiving 20 responses.



Analyse Holiday Study Patterns

Examine how many hours students study daily during holidays and how they allocate free time.



Correlate Habits with CGPA

Compare CGPA distribution against study routines, time-table discipline, and attendance views.



Assess AI Tool Adoption

Understand frequency of AI tool (ChatGPT, Gemini) usage for academic learning.

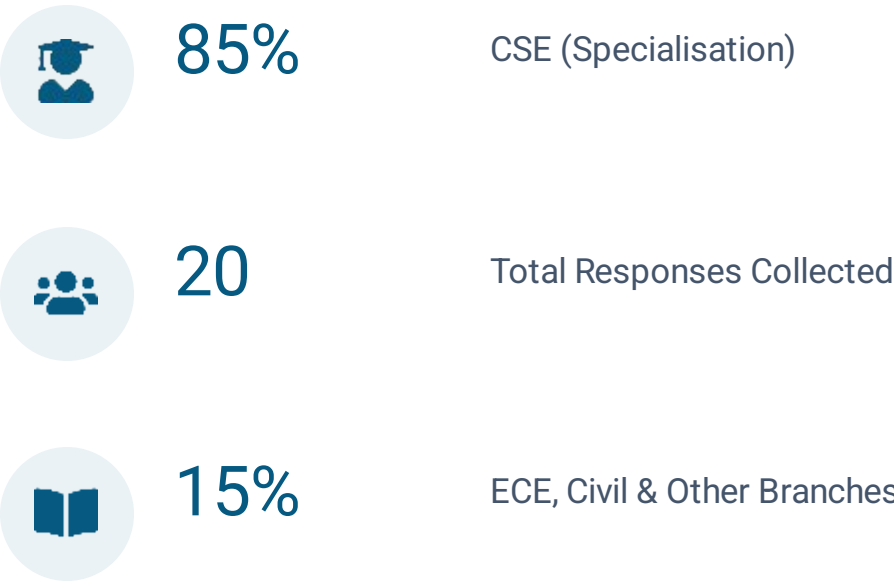


Identify Key Challenges

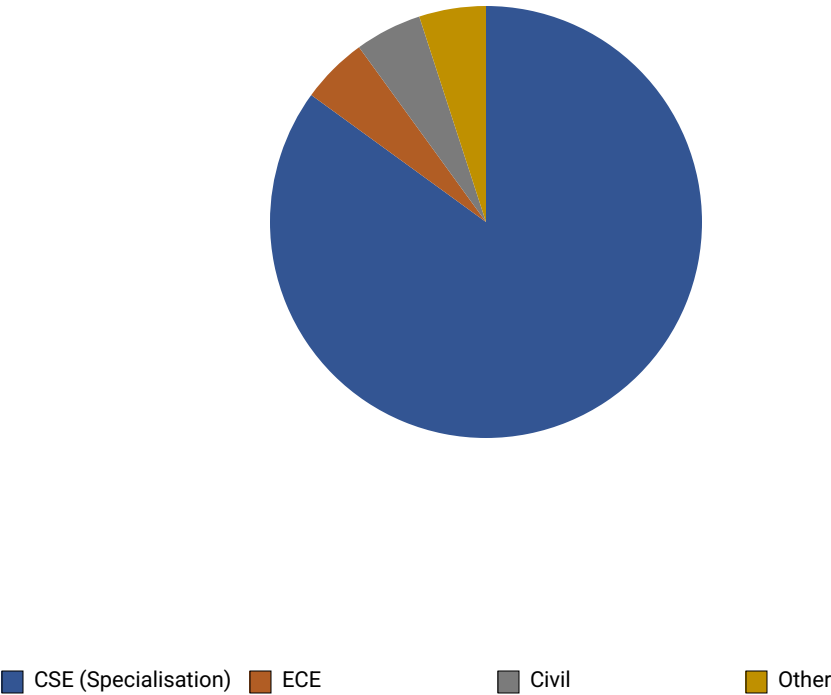
Pinpoint the biggest obstacles students face — time management, motivation, distractions.

Respondent Profile

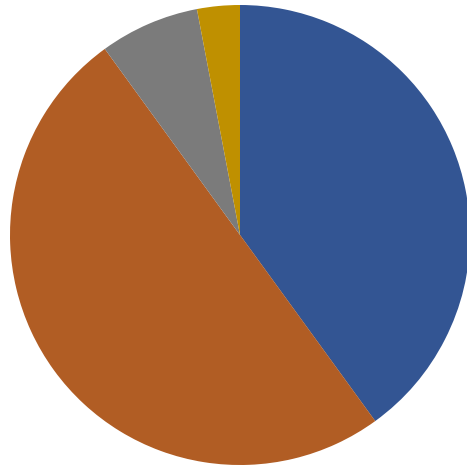
All 20 respondents were engineering students, with the overwhelming majority pursuing a specialised track within Computer Science Engineering. This concentration means the findings are most representative of CS-specialisation students, with limited representation from ECE, Civil, and other branches.



Branch Distribution



Daily Study Hours During Holidays



■ Less than 1 hour ■ 1-2 hours ■ 2-4 hours ■ More than 4 hours

90% of students study 2 hours or less per day

40% study less than 1 hour daily, and 50% study 1-2 hours. Only a small minority (roughly 1 in 10) push beyond 2 hours of daily study during holiday breaks.

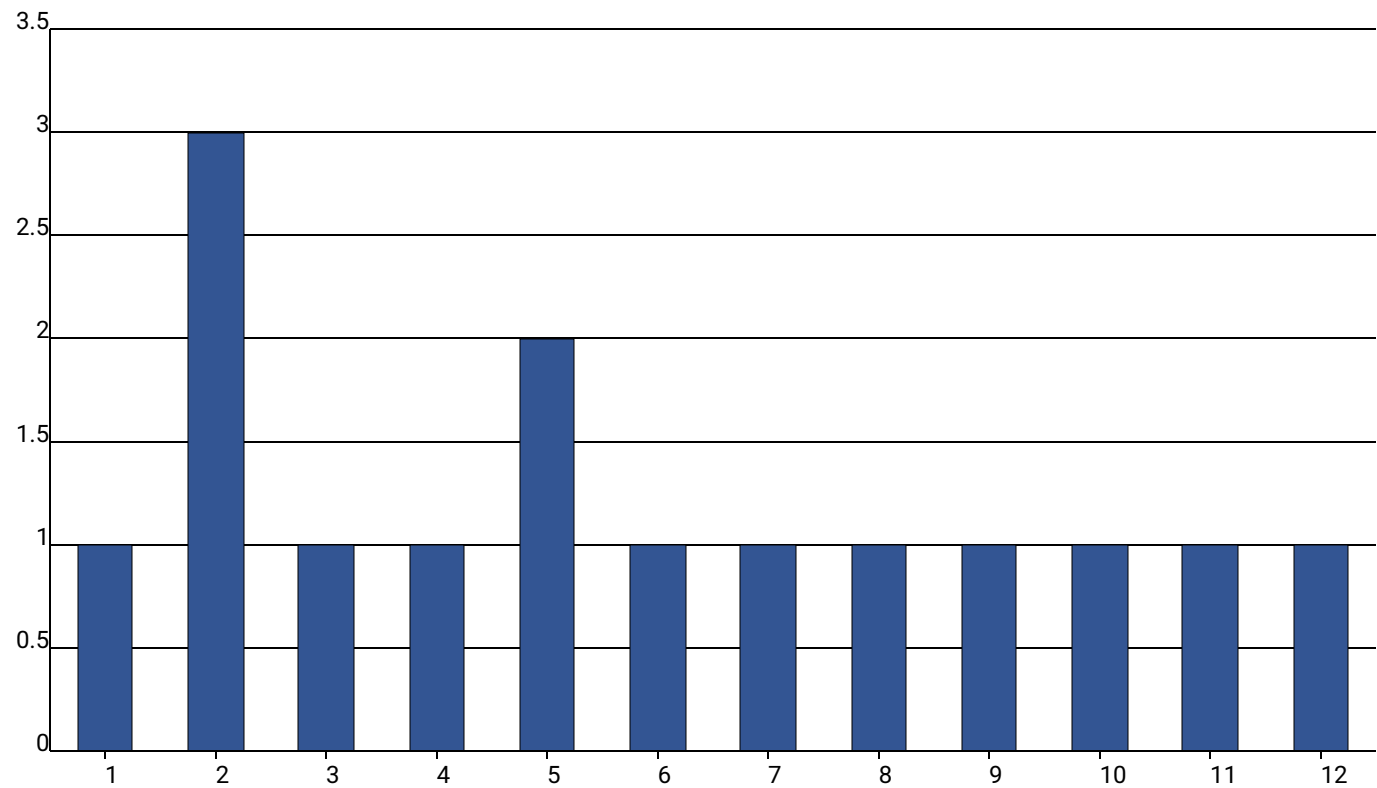
Holiday study is light-touch, not intensive

Very few students (about 10%) treat holidays as a period for extended (2-4 hours or more) daily study, suggesting holidays are largely viewed as a break from intensive academics.

Implication

Even small, consistent gains in holiday study time (e.g., moving from "less than 1 hour" to "1-2 hours") could shift the bulk of the cohort toward more productive routines.

Current CGPA Distribution



Wide Spread, CGPA 6 to 9.55

CGPA values range from 6.0 to 9.55 across the 19 students who reported, indicating a highly diverse academic standing within the same cohort.

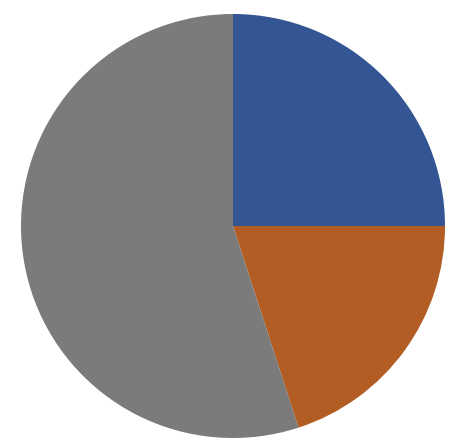


CGPA 7.0 is the Most Common

The most frequent value is 7.0 (15.8% of students), followed by 8.2 (10.5%). The remaining values are each held by a single student, reflecting an otherwise evenly spread distribution.

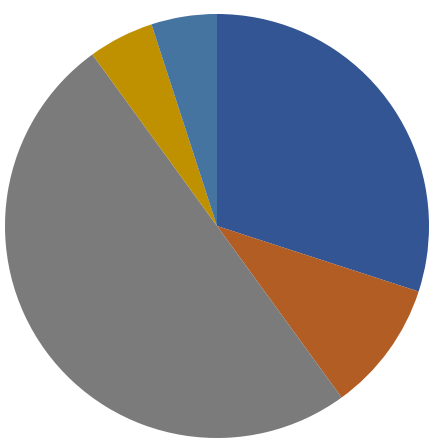
Perceived Impact on Academic Performance

Do holidays help improve academic performance?



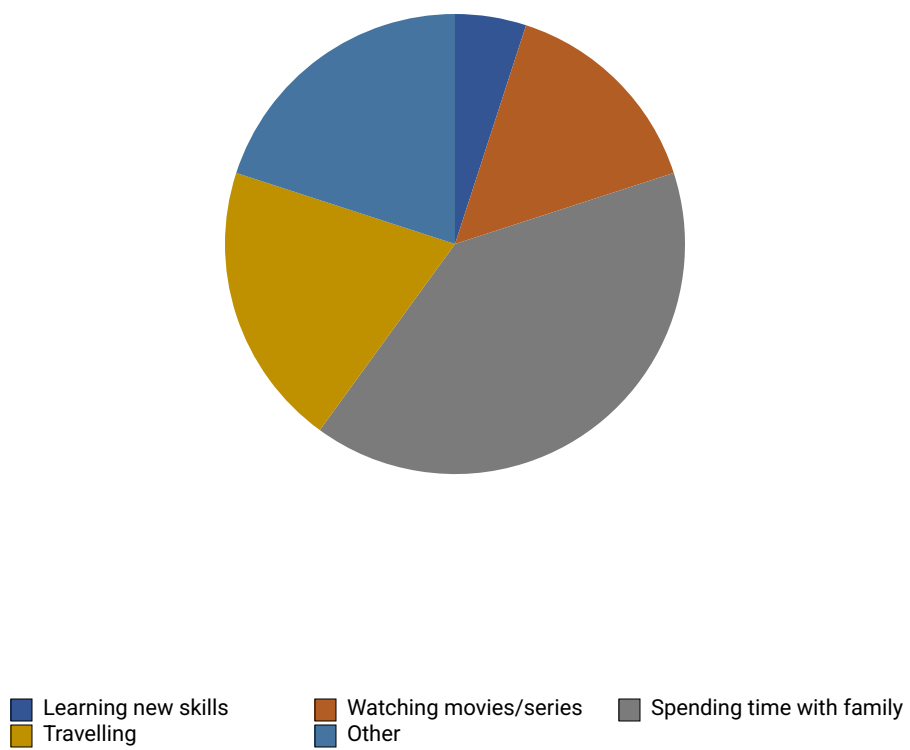
■ Yes ■ No ■ Sometimes

Does attendance affect academic performance?



■ Strongly Agree ■ Agree ■ Neutral ■ Disagree ■ Strongly Disagree

How Students Spend Their Holidays



Family Time Dominates (40%)

Spending time with family is the single largest use of holiday time, far ahead of any other activity.



Travel & Other Activities (40% combined)

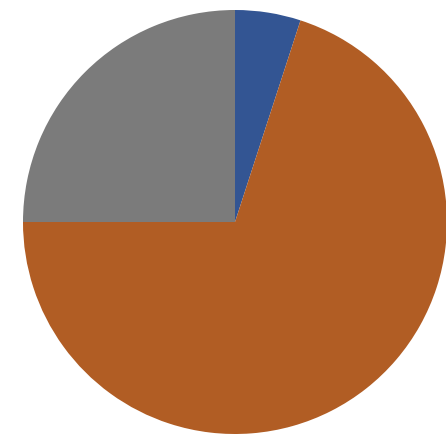
20% travel and 20% report other activities, alongside 15% watching movies/series and 5% learning new skills.



Dedicated 'Studying' Was Not Selected as Primary Use

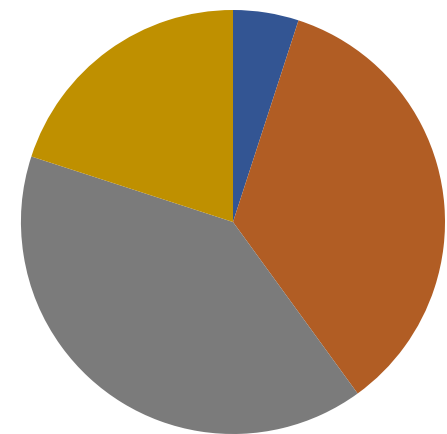
Routine Discipline & Past Academic Standing

Do you follow a time table?



Always Sometimes Never

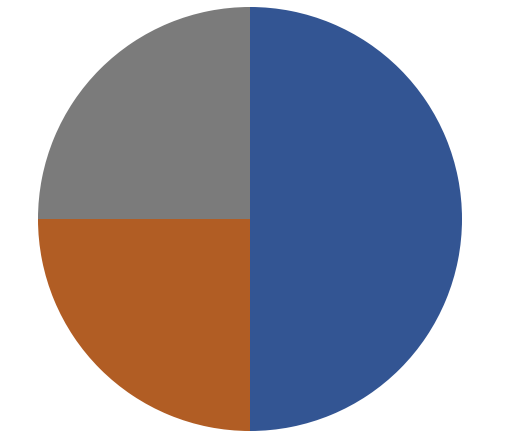
Percentage maintained in previous class



Below 75% 75% - 85% 85% - 95% Above 95%

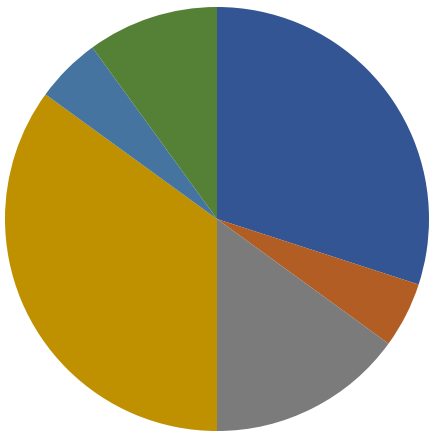
AI Tool Usage & Biggest Study Challenges

Do you use AI tools (ChatGPT, Gemini, etc.) for learning?



■ Daily ■ Weekly ■ Occasionally

What is your biggest challenge in studies?



■ Time Management ■ Lack of Motivation ■ Difficulty Understanding Subjects
■ Excessive Phone/Social Media ■ Family Responsibilities ■ Other

Conclusions



Holiday study time is low but improvable

90% study 2 hours or less daily during holidays; even modest increases could shift outcomes for the majority.



Digital distraction is the top barrier

Excessive phone/social media (35%) and time management (30%) are the leading challenges - tackling these could unlock the largest gains.



Routine discipline is weak

Only 5% always follow a time table; structured routines, even loosely enforced, may help convert free time into productive study.



AI tools are widely adopted

100% of students use AI tools at least occasionally, with 50% daily - a strong foundation for AI-assisted, self-paced holiday learning plans.



Strong academic base, wide CGPA spread

75% previously scored above 85%, yet current CGPA spans 6.0-9.55 - habit changes (not just ability) likely drive the spread.



Attendance attitudes need clarity

50% are neutral on whether attendance affects performance - clearer communication of its impact could improve engagement.